

ECEF Navigation Update system

Navigation Equation $\underline{a}_{ib}^e = \underline{f}_{ib}^e + \gamma_{ib}^e(\underline{r}_{ib}^e)$

What we wanna get $\underline{v}_{eb}^e$

$$\underline{r}_{eb}^e = \underline{r}_{ib}^e - \underline{r}_{ie}^e = \underline{r}_{ib}^e$$

$$\Rightarrow \dot{\underline{r}}_{eb}^e = \underline{v}_{eb}^e = \dot{\underline{r}}_{ib}^e \Rightarrow \dot{\underline{v}}_{eb}^e = \ddot{\underline{r}}_{ib}^e$$

We know:

$$\ddot{\underline{r}}_{ib}^e = -(\underline{\Omega}_{ie}^e \underline{\Omega}_{ie}^e + \dot{\underline{\Omega}}_{ie}^e) \underline{r}_{ib}^e - 2 \underline{\Omega}_{ie}^e \dot{\underline{r}}_{ib}^e + \underline{a}_{ib}^e$$

$$\dot{\underline{v}}_{eb}^e$$

$$\begin{aligned} \Rightarrow \dot{\underline{v}}_{eb}^e &= \underline{f}_{ib}^e + \gamma_{ib}^e(\underline{r}_{ib}^e) - (\underline{\Omega}_{ie}^e \underline{\Omega}_{ie}^e + \dot{\underline{\Omega}}_{ie}^e) \underline{r}_{ib}^e - 2 \underline{\Omega}_{ie}^e \dot{\underline{r}}_{ib}^e \\ &= \underline{C}_b^e \underline{f}_{ib}^b + \underline{g}_b^e(\underline{r}_{eb}^e) - 2 \underline{\Omega}_{ie}^e \underline{v}_{eb}^e \end{aligned}$$

And we need to update $\underline{C}_b^e$

$$\dot{\underline{C}}_b^e = \underline{C}_b^e \cdot \underline{\Omega}_{eb}^b$$

$$\underline{\omega}_{eb}^b = \underline{\omega}_{ib}^b - \underline{\omega}_{ie}^b \Rightarrow \underline{\Omega}_{eb}^b = \underline{\Omega}_{ib}^b - \underline{\Omega}_{ie}^b$$

$$\dot{\underline{C}}_b^e = \underline{C}_b^e \underline{\Omega}_{ib}^b - \underline{C}_b^e \underline{\Omega}_{ie}^b$$

$$\Rightarrow \dot{\underline{C}}_b^e = \underline{C}_b^e \underline{\Omega}_{ib}^b - \underline{\Omega}_{ie}^e \underline{C}_b^e$$



